Fields and Galois theory

Chapter 1. Polynomial rings

Chapter 2. Field extensions

§2.1 Field homomorphisms and isomorphisms

Definition 1.1. Let K and L be fields. A **field homomorphism** from K to L is a map $\sigma : K \to L$ that preserves the field operations, that is, for all $a, b \in K$, we have

- $\sigma(a + b) = \sigma(a) + \sigma(b)$,
- $\sigma(ab) = \sigma(a)\sigma(b)$,
- $\sigma(1_K) = 1_L$, where 1_K and 1_L denote the multiplicative identities of K and L, respectively.

A homomorphism $\sigma : K \to L$ is called an **isomorphism** if there exists a homomorphism $\tau : L \to K$ such that $\tau \circ \sigma = \mathrm{id}_K$ and $\sigma \circ \tau = \mathrm{id}_L$, where id_K and id_L denote the identity maps on K and L, respectively.

Exercise 1.2. Show that if $\sigma : K \to L$ is a field homomorphism, then σ is injective. Is it true that σ is surjective? If yes, prove it. If not, give a counterexample.

Exercise 1.3. Let $\sigma : K \to L$ be a field homomorphism. Show that σ is an isomorphism if and only if σ is surjective.

§2.2 Extension of fields

For any field k, the ring of polynomials in one variable x over k is denoted by $k[x]$. The field of rational functions in one variable x over k is denoted by $k(x)$, and it consists of all elements of the form $\frac{f(x)}{g(x)}$, where $f(x), g(x) \in k[x]$ and $g(x) \neq 0$. The field $k(x)$ is the field of fractions of the integral domain $k[x]$.

Definition 1.4. Let K be a field. A **field extension** of K is a field L that contains K as a subfield. We often denote a field extension by L/K, which indicates that L is an extension of K. The field K is called the **base field**, and the field L is called the **extension field**.

Example 1.5.

- The field of complex numbers $\mathbb{C}$ is an extension of the field of real numbers $\mathbb{R}$.
- Similarly, the field of real numbers $\mathbb{R}$ is an extension of the field of rational numbers $\mathbb{Q}$.
- For any field k, the field of rational functions $k(x)$ in one variable x over k is an extension of k.
- $k((x))$

Definition 1.6. Let L/K be a field extension. An **intermediate extension** of L/K is a field M such that $K \subseteq M \subseteq L$. In this case, we say that M/K is a subextension of L/K.

Exercise 1.7. Show that any element of $k(t)$ is a root of some quadratic polynomial with coefficients in $k(t^2)$.

§2.3 Generated subfields

Definition 1.8. Let L/K be a field extension. Let S be a subset of L. The **field generated by S over K**, denoted by $K(S)$, is defined to be the smallest subfield of L that contains both K and S. We say that the field extension L/K is **generated by S** if $L = K(S)$. The **ring generated by S over K**, denoted by $K[S]$, is defined to be the smallest subring of L that contains both K and S.

Remark 1.9. Note that the field generated by S over K is the intersection of all subfields of L that contain both K and S. Similarly, the ring generated by S over K is the intersection of all subrings of L that contain both K and S.

Definition 1.10. Let L/K be a field extension. We say that the extension L/K is **finitely generated** if there exists a finite subset $S \subset L$ such that $L = K(S)$.

Exercise 1.11. Let k be a field. Consider the field extension $k(x_1, x_2, ..., x_n)/k$, where $k(x_1, x_2, ..., x_n)$ is the field of rational functions in n variables $x_1, x_2, ..., x_n$ over k. Show that $k(x_1, x_2, ..., x_n)$ is generated by the elements $x_1, x_2, ..., x_n$ over k, that is, show that $k(x_1, x_2, ..., x_n) = k(\{x_1, x_2, ..., x_n\})$. Also show that the ring $k[x_1, ..., x_n]$ of polynomials in n variables $x_1, ..., x_n$ over k is generated by the elements $x_1, ..., x_n$ over k, that is, show that $k[x_1, ..., x_n] = k[\{x_1, ..., x_n\}]$.

Remark 1.12. The field of rational functions $k(x_1, x_2, ..., x_n)$ in n variables $x_1, x_2, ..., x_n$ over k is the field of fractions of the integral domain $k[x_1, ..., x_n]$ of polynomials in n variables $x_1, ..., x_n$ over k. Moreover, the field $k(x_1, x_2, ..., x_n)$ is generated by the elements $x_1, x_2, ..., x_n$ over k, and hence, the extension $k(x_1, x_2, ..., x_n)/k$ is finitely generated. Furthermore, the ring $k[x_1, ..., x_n]$ is generated by the elements $x_1, ..., x_n$ over k.

Exercise 1.13. Let L/K be a field extension, and let S be a subset of L. Show that the field generated by S over K is equal to the set of all elements of L that can be expressed as a rational function in the elements of S with coefficients in K. In other words, show that

$$K(S) = \bigcup_{\substack{S_f \subset S, \\ |S_f| < \infty}} K(S_f).$$

Also show that the ring generated by S over K is equal to the set of all elements of L that can be expressed as a polynomial in the elements of S with coefficients in K. In other words, show that

$$K[S] = \bigcup_{\substack{S_f \subset S, \\ |S_f| < \infty}} K[S_f].$$

Definition 1.14. Let L/K be a field extension, and let $\alpha_1, \alpha_2, ..., \alpha_n \in L$. Then the field $K(\{\alpha_1, \alpha_2, ..., \alpha_n\})$ is denoted by $K(\alpha_1, \alpha_2, ..., \alpha_n)$.

Exercise 1.15. Show that the fields $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3})$ are not isomorphic. What can be said about the fields $\mathbb{Q}(\sqrt{2}, \sqrt{6}), \mathbb{Q}(\sqrt{3}, \sqrt{6})$?

Exercise 1.16. Show that $\mathbb{Q}\left(\sqrt{2}\right)$ is equal to $\mathbb{Q}\left[\left\{\sqrt{2}\right\}\right]$.

Exercise 1.17. Show that $\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right)$ is equal to $\mathbb{Q}\left(\sqrt{2}+\sqrt{3}\right)$.

Exercise 1.18. Show that $\mathbb{Q}\left(\sqrt{2}, \omega\right)$ is equal to $\mathbb{Q}\left(\sqrt{2}+\omega\right)$, where ω denotes a primitive cube root of unity in $\mathbb{C}$.

§2.4 Degree of field extensions

If L/K is a field extension, then L can be viewed as a vector space over K, where scalar multiplication is given by the field multiplication. In particular, we can consider the dimension of L as a vector space over K.

Exercise 1.19. Show that the map

$$\mathbb{C} \times \mathbb{C} \to \mathbb{C}, (a, b) \mapsto \overline{a}b,$$

where $\overline{a}$ denotes the complex conjugate of a, defines a $\mathbb{C}$-linear structure on $\mathbb{C}$ that makes $\mathbb{C}$ into a vector space over $\mathbb{C}$.

Definition 1.20. Let L/K be a field extension. We say that L is a **finite extension** of K if L is a finite-dimensional vector space over K. In this case, we define the **degree** of the extension L/K to be the dimension of L as a vector space over K, and we denote it by $[L : K]$. If L is not finite-dimensional over K, we say that the extension is **infinite**.

Exercise 1.21. Let L/K be a field extension. Show that the degree of the extension L/K is equal to 1 if and only if $L = K$.

Exercise 1.22. Show that $[k(x) : k] = \infty$ for any field k.

Exercise 1.23. Show that $k(t)/k(t^2)$ is a finite extension of degree 2.

Exercise 1.24. Determine the degrees of the following extensions.

$$\mathbb{Q}\left(\sqrt{2}\right)/\mathbb{Q}, \mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right)/\mathbb{Q}, \mathbb{Q}\left(\sqrt{2}, \sqrt[3]{3}\right)/\mathbb{Q}, \mathbb{Q}\left(\sqrt{2}, \sqrt{3}, \sqrt{5}\right)/\mathbb{Q}.$$

Exercise 1.25. Determine the degrees of the following extensions over $\mathbb{Q}$.

$$\mathbb{Q}(i), \mathbb{Q}(\omega), \mathbb{Q}(\zeta_n).$$

Exercise 1.26. Let L/K be a field extension. Show that if L/K is a finite extension of degree n, then any element of L is a root of a polynomial of degree at most n with coefficients in K.

Now let's look at some examples of finite extensions.

Example 1.27. Consider the field extension $\mathbb{Q}\left(\sqrt{2}\right)/\mathbb{Q}$. We claim that this is a finite extension of degree 2. To see this, note that any element of $\mathbb{Q}\left(\sqrt{2}\right)$ can be written in the form $a + b\sqrt{2}$ for some

$a, b \in \mathbb{Q}$. The set $B = \left\{1, \sqrt{2}\right\}$ is linearly independent over $\mathbb{Q}$, and it spans $\mathbb{Q}\left(\sqrt{2}\right)$, so it forms a basis. Therefore, the dimension of $\mathbb{Q}\left(\sqrt{2}\right)$ as a vector space over $\mathbb{Q}$ is 2, and we have $\left[\mathbb{Q}\left(\sqrt{2}\right) : \mathbb{Q}\right] = 2$.

Lemma 1.28. Let L/K, K/F be field extensions. If $L/K, K/F$ are finite extensions, then so is L/F, and the degrees of these extensions satisfy

$$[L : F] = [L : K][K : F].$$

Proof. Note that if the degree of the extension K/F is infinite, then observing that K is an F-subspace of L, it follows that the degree of the extension L/F is also infinite. Also note that if the degree of the extension L/K is infinite, then the degree of the extension L/F is also infinite, since any subset of L that generates L as a K-vector space also generates L as an F-vector space. This shows that if either $[L : K]$ or $[K : F]$ is infinite, then $[L : F]$ is also infinite.

Let $\{u_1, u_2, ..., u_m\}$ be a basis of L over K, and let $\{v_1, v_2, ..., v_n\}$ be a basis of K over F. We claim that the set

$$B = \left\{u_i v_j : 1 \leq i \leq m, 1 \leq j \leq n\right\}$$

is a basis of L over F. Indeed, for any $\alpha \in L$, we can write

$$\alpha = \sum_{i=1}^{m} c_i u_i$$

for some $c_1, ..., c_m$ lying in K. Moreover, for each c_i, we can write

$$c_i = \sum_{j=1}^{n} d_{ij} v_j$$

for some d_{ij} lying in F. This shows that

$$\alpha = \sum_{i=1}^{m} \sum_{j=1}^{n} d_{ij} u_i v_j,$$

which shows that B generates L over F. To show that B is linearly independent over F, suppose that we have a linear combination

$$\sum_{i=1}^{m} \sum_{j=1}^{n} e_{ij} u_i v_j = 0$$

for some e_{ij} lying in F. Then we can rewrite this as

$$\sum_{i=1}^{m} \left(\sum_{j=1}^{n} e_{ij} v_j \right) u_i = 0.$$

Since $\{u_1, u_2, ..., u_m\}$ is linearly independent over K, we must have

$$\sum_{j=1}^{n} e_{ij} v_j = 0$$

for each $i = 1, ..., m$. Since $\{v_1, v_2, ..., v_n\}$ is linearly independent over F, we must have $e_{ij} = 0$ for all i, j. This shows that B is linearly independent over F, and hence it is a basis of L over F. Therefore, we have

$$[L : F] = |B| = mn = [L : K][K : F].$$

□

Exercise 1.29. If an extension has prime degree, show that it has no proper intermediate extensions.

Lemma 1.30. Let L/K be a field extension, and let $\alpha_1, \alpha_2, ..., \alpha_n \in L$. Then the extension $K(\alpha_1, \alpha_2, ..., \alpha_n)/K$ is finite if and only if each of the extensions $K(\alpha_i)/K$ is finite for $i = 1, ..., n$. Moreover, if these extensions are finite, then we have

$$[K(\alpha_1, \alpha_2, ..., \alpha_n) : K] \leq \prod_{i=1}^{n} [K(\alpha_i) : K].$$

§2.5 Algebraic elements and algebraic extensions

Exercise 1.31. Let L/K be a field extension, and let $\alpha \in L$.

- Show that the multiplication by α map, given by

$$L \to L, \quad x \mapsto \alpha x,$$

  defines a K-linear transformation of L as a vector space over K.
- If L/K is a finite extension, the characteristic polynomial of this linear transformation is a monic polynomial in $K[x]$ of degree $[L : K]$ vanishing at α.
- Assume that α is algebraic over K, and $L = K(\alpha)$. Prove that the characteristic polynomial of this linear transformation is equal to the minimal polynomial of α over K.

Definition 1.32. Let L/K be a field extension. An element $\alpha \in L$ is said to be **algebraic** over K if there exists a non-zero polynomial $f(x) \in K[x]$ such that $f(\alpha) = 0$. If no such polynomial exists, we say that α is **transcendental** over K.

Exercise 1.33. Is it true that algebraic extensions are finitely generated? If yes, prove it. If not, give a counterexample.

Corollary 1.34. Let L/K be a field extension, and let α be an element of L. Then the following statements are equivalent.

- The extension $K(\alpha)/K$ is finite.
- The element α is algebraic over K.

Moreover, α is algebraic over K if and only if α^n is algebraic over K for some positive integer n.

Definition 1.35. Let L/K be a field extension. If an element $\alpha \in L$ is algebraic over K, then the **minimal polynomial** of α over K is the unique monic polynomial $f(x) \in K[x]$ of least degree such that $f(\alpha) = 0$.

Lemma 1.36. Let L/K be a field extension, and let $\alpha \in L$ be algebraic over K. Then the following statements hold.

- The minimal polynomial of α over K is irreducible in $K[x]$.
- If $f(x) \in K[x]$ is any polynomial such that $f(\alpha) = 0$, then the minimal polynomial of α over K divides $f(x)$ in $K[x]$.
- The degree of the extension $K(\alpha)/K$ is equal to the degree of the minimal polynomial of α over K.

- We have

$$K(\alpha) = K[\alpha].$$

Lemma 1.37 (Finite extensions are algebraic). Let L/K be a field extension. If L/K is a finite extension, then L is algebraic over K, and it is finitely generated as a field extension of K.

Proposition 1.38. Let L/K be a field extension, and let X be a subset of L. Assume that every element of X is algebraic over K. Then the extension $K(X)/K$ is algebraic. Moreover, if X is finite, then the extension $K(X)/K$ is finite.

Theorem 1.39. Let L/K, K/F be field extensions. If L/K, K/F are algebraic extensions, then so is L/F.

Lemma 1.40. Let L/K be a field extension. The set of all elements of L that are algebraic over K forms a subfield of L, and it is the largest algebraic extension of K contained in L.

Definition 1.41. Let L/K be a field extension. The **algebraic closure** of K in L is the set of all elements of L that are algebraic over K. The field extension L/K is called an **algebraic extension** if every element of L is algebraic over K.

Exercise 1.42. Let $\overline{\mathbb{Q}}$ denote the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$. Show that $\overline{\mathbb{Q}}$ contains an extension of $\mathbb{Q}$ of degree n for every positive integer n.

§2.6 Compositum of field extensions

Definition 1.43. Let L_1/F and L_2/F be field extensions, contained in a common extension K of F. The **compositum** of L_1 and L_2 over F, denoted by L_1L_2, is defined to be the smallest subfield of K that contains both L_1 and L_2. In other words, L_1L_2 is the intersection of all subfields of K that contain both L_1 and L_2. The compositum L_1L_2 is also called the **composite field** of L_1 and L_2 over F.

Exercise 1.44. If L_1/F and L_2/F are finite extensions contained in a common extension K of F, and if

$$L_1 = F(\alpha_1, \alpha_2, ..., \alpha_m), \quad L_2 = F(\beta_1, \beta_2, ..., \beta_n),$$

then show that

$$L_1L_2 = F(\alpha_1, \alpha_2, ..., \alpha_m, \beta_1, \beta_2, ..., \beta_n).$$

Exercise 1.45. Show that if L_1/F and L_2/F are finite extensions contained in a common extension K of F, then the compositum L_1L_2/F is also a finite extension of F, and we have

$$[L_1L_2 : F] \leq [L_1 : F][L_2 : F].$$

Moreover, if $[L_1 : F]$ and $[L_2 : F]$ are coprime, then we have

$$[L_1L_2 : F] = [L_1 : F][L_2 : F].$$